

PROOF TECHNIQUES

CSE 2213 – Discrete Mathematics

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WHAT TO PROVE? HOW TO PROVE?

Prove that, if an integer n is even, then $3n + 2$ is also even.

- n even, so $3n$ also even
- $3n$ even, so $3n + 2$ even

CAN WE PROVE THE REVERSE?

Prove that, if n is an integer, and $3n + 2$ is even, then n is also even.

- $3n + 2$ even, so $3n$ is also even
- We know, even $\times$ even = even
odd $\times$ even = even
odd $\times$ odd = odd
- As $3n$ is even and 3 is odd, so n must be even

PROOF TECHNIQUES

Three basic techniques –

- ? Direct proof
- ? Proof by contradiction
- ? Proof by contraposition

DIRECT PROOF

To prove $p \rightarrow q$, we first assume that p is true, and hence prove that q is true

- We reach the conclusion directly from our assumption

EXAMPLE

Prove that the product of two rational numbers is rational

Direct Proof:

Suppose that x and y are both rational.

- Then there are integers m and n so that $x = m/n$, and there are integers p and q so that $y = p/q$.
- Multiplying, we can conclude $xy = (mp)/(nq)$.
- That exhibits xy as a quotient of integers. Therefore the product xy is also rational.

PROOF BY CONTRAPOSITION

- To prove $p \rightarrow q$, we first assume that $\neg q$ is true, and hence prove that $\neg p$ is true
 - We actually prove the contrapositive of the actual sentence, i.e. $\neg q \rightarrow \neg p$
- What is the difference between proof by contraposition and proof by contradiction?

PROOF BY CONTRADICTION

To prove $p \rightarrow q$, we first assume that $p \rightarrow \neg q$, and start logical arguments to reach a conclusion that contradicts our assumption

- We actually show that if $p \rightarrow q$ is false, the sun rises in the west :P

EXAMPLE

Prove that $\sqrt{2}$ is an irrational number.

Prove this using contradiction

Homework

The Number of primes is infinite

Prove this using contradiction



EXAMPLE

Prove that if n is an integer and $n^3 + 5$ is odd, then n is even.

Proof By Contraposition

The contrapositive is “If n is odd, then $n^3 + 5$ is even.” Assume that n is odd. We can now write $n = 2k + 1$ for some integer k . Then $n^3 + 5 = (2k + 1)^3 + 5 = 8k^3 + 12k^2 + 6k + 6 = 2(4k^3 + 6k^2 + 3k + 3)$. Thus $n^3 + 5$ is two times some integer, so it is even by the definition of an even integer.

VACUOUS AND TRIVIAL PROOFS

If we prove a conditional statement by disproving its hypothesis, then the proof technique is called vacuous proof

If we prove a conditional statement by proving its conclusion, then the proof technique is called trivial proof

VACUOUS PROOF

Prove that if n is an integer with $10 \leq n \leq 15$ which is a perfect square, then n is also a perfect cube.

Solution: Note that there are no perfect squares n with $10 \leq n \leq 15$, because $3^2 = 9$ and $4^2 = 16$. Hence, the statement that n is an integer with $10 \leq n \leq 15$ which is a perfect square is false for all integers n . Consequently, the statement to be proved is true for all integers n . 

TRIVIAL PROOF

Let $P(n)$ be “If a and b are positive integers with $a \geq b$, then $a^n \geq b^n$,” where the domain consists of all nonnegative integers. Show that $P(0)$ is true.

Solution: The proposition $P(0)$ is “If $a \geq b$, then $a^0 \geq b^0$.” Because $a^0 = b^0 = 1$, the conclusion of the conditional statement “If $a \geq b$, then $a^0 \geq b^0$ ” is true. Hence, this conditional statement, which is $P(0)$, is true. This is an example of a trivial proof. Note that the hypothesis, which is the statement “ $a \geq b$,” was not needed in this proof. 

QUICK EXERCISE

- To prove that $\angle AOC + \angle BOC = 180^\circ$
- We can see that $\angle AOC + \angle BOC = \angle AOD + \angle COD + \angle BOC = \angle AOD + \angle BOD = 90^\circ + 90^\circ = 180^\circ$
- What was the technique of this proof?

PROVING EQUIVALENCES

- Prove that “For every positive integer n , n is odd if and only if n^2 is odd”
 - First prove “if n is odd then n^2 is odd”
 - Then prove “if n^2 is odd then n is odd”

DISPROVING UNIVERSALLY QUANTIFIED PROPOSITIONS

Prove that “**Every** positive integer is the sum of the squares of two integers” is false.

? We just find a **counterexample**, that invalidates the use of “every” in the sentence

? Here, a counterexample is 3 (because $3 = 1 + 2 = 0 + 3$, and 2 and 3 are not squares of any integer)

STANDARD FORMS OF DIFFERENT NUMBERS

Type	Standard form
Even number	$2k$
Odd number	$2k + 1$
Perfect square	k^2
Rational number	a/b where, $b \neq 0$ and a and b have no common factors
Positive fraction	a/b where, $b \neq 0$ and a and b both have same sign

EXAMPLE

- To prove $p \rightarrow q$, we first assume that p is true, and hence prove that q is true
 - We reach the conclusion directly from our assumption
- The proof of উপপাদ্য ১ is a direct proof

EXAMPLE

Prove that, if x is irrational, then $\frac{1}{x}$ is also irrational

We will use proof by contraposition. The contrapositive is “If $1/x$ is rational, then x is rational.” Suppose that $1/x$ is rational and $x \neq 0$. Then there exists integers p and q such that $1/x = p/q$ and $q \neq 0$. $\frac{1}{x} \neq 0$ because $1 \neq x \cdot 0$, this would mean that $p \neq 0$. Since $p \neq 0$, then $x = \frac{1}{\frac{1}{x}} = \frac{1}{\frac{p}{q}} = \frac{q}{p}$. Hence x can be written as a quotient of two integers with a nonzero denominator. Thus, x is rational.

EXAMPLE

Prove that, the sum of a rational and an irrational number is irrational

Assume that a is rational, b is irrational, and $a + b$ is rational. Since a and $a + b$ are rational, we can write them as fractions.

$$\text{Let } a = \frac{c}{d} \text{ and } a + b = \frac{m}{n}$$

Substituting $a = \frac{c}{d}$ in $a + b = \frac{m}{n}$ gives the following:

$$\frac{c}{d} + b = \frac{m}{n}$$

Now, let's subtract $\frac{c}{d}$ from both sides of the equation.

$$b = \frac{m}{n} - \frac{c}{d}, \text{ or}$$

$$b = \frac{m}{n} + \left(-\frac{c}{d}\right)$$

Since the rational numbers are closed under addition, $b = \frac{m}{n} + \left(-\frac{c}{d}\right)$ is a rational number. However, the assumptions said that b is irrational, and b cannot be both rational and irrational. This is our contradiction, so it must be the case that the sum of a rational and an irrational number is always irrational.

QUICK EXERCISE

Prove or disprove that the product of two irrational numbers is irrational

Can we find an example to disprove the statement?